

Module 16: History of Life — Questions

Introduction

The final module traces the billions of years of biological history on Earth. These questions will test your grasp of the chronological milestones of evolutionary history, the catastrophic resets of mass extinctions, and the principles of reading phylogenetic trees.

Practice Questions

1. Approximately how old is the Earth, and how quickly after its formation did the very first single-celled life (prokaryotes) emerge?
2. What was the "Oxygen Revolution" and what biological process was primarily responsible for introducing massive amounts of oxygen into Earth's early atmosphere?
3. Place the following milestones in sequential chronological order: Eukaryotes, Multicellular Life, Prokaryotes, Oxygenification, Colonization of Land.
4. What is a mass extinction, as defined by the fossil record?
5. How many major mass extinction events have significantly altered the trajectory of life on Earth?
6. What is an adaptive radiation, and why does it frequently follow in the immediate aftermath of a mass extinction?
7. By evaluating the fossil record, what makes the Permian mass extinction particularly notable?
8. To what event do scientists largely attribute the Cretaceous mass extinction (the death of the non-avian dinosaurs)?
9. What is phylogeny?
10. When looking at a phylogenetic tree (cladogram), what does a "branch point" or "node" physically represent?
11. Define what makes a biological group a true "clade" (a monophyletic group).
12. What is a shared derived character, and how is it used to construct an evolutionary tree?
13. Contrast a shared derived character with a shared ancestral character.

14. Why do phylogenetic trees sometimes undergo massive revisions when new DNA sequence evidence is discovered?
15. If two organisms are found located directly adjacent to one another on a terminal branch of a phylogenetic tree, what does that signify about their common ancestry relative to other organisms on the tree?